

Olympiad Paper I.

1. Let n be a positive integer. Prove that $z^{n+1} - z^n - 1 = 0$ has a root satisfying $|z| = 1$ if and only if $n + 2$ is divisible by 6.
2. An equilateral triangle ABC of side n is divided into n^2 equilateral triangles of side 1 by lines parallel to the sides of ABC . Each point which is a vertex of at least one of these unit triangles is labeled with a real number. The points A, B and C are labeled with a, b and c respectively. In each rhombus composed of two unit triangles with a common side, the sums of the labels on the two sets of opposite vertices are equal.
 - (a) Determine the shortest distance between a point with the largest label and a point with the smallest label.
 - (b) Determine the sum of all labels.
3. In a tournament, every two players have exactly one game between them. There are no ties. A player A is awarded a prize if for every other player B , either A defeats B or A defeats some player C who defeats B . Prove that if only one player is awarded a prize, then this player defeats every other player.

Olympiad Paper II.

1. Given are five points inside an equilateral triangle of area 1. Prove that there exist three equilateral triangles such that they are homothetic to the given triangle, each of the five points is inside at least one of these three triangles, and their combined area is at most 0.64.
2. A tetrahedron is such that a sphere centred at a point O is tangent to all six of its sides. Moreover, four spheres with the vertices as their respective centres are externally tangent to one another, and all are tangent to another sphere with centre O . Prove that the tetrahedron is regular.
3. The sum of m distinct positive even integers and n distinct positive odd integers is 1987. What is the maximum value of $3m + 4n$?

Olympiad Paper I.

1. Let $a_1, a_2, \dots, a_n$ be given real numbers, not all being 0.

The real numbers $r_1, r_2, \dots, r_n$ are such that for all real numbers $x_1, x_2, \dots, x_n$,

$$r_1(x_1 - a_1) + r_2(x_2 - a_2) + \dots + r_n(x_n - a_n)$$

is less than or equal to

$$\sqrt{x_1^2 + x_2^2 + \dots + x_n^2} - \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}.$$

Determine $r_1, r_2, \dots, r_n$.

2. Given are two concentric circles one having twice the radius of the other. Inscribed in the smaller circle is a convex quadrilateral $ABCD$. The extensions of AB, BC, CD and DA intersect the larger circle at C_1, D_1, A_1 and B_1 respectively.
- (a) Prove that the perimeter of $A_1B_1C_1D_1$ is not less than twice that of $ABCD$.
 - (b) Determine when equality holds.
3. Given is a sequence of n real numbers. A dragon is defined to be a block of consecutive terms whose average is greater than 1988, and the head of this dragon is the first term in the block. A single term greater than 1988 is also considered to be a dragon as well as its head. Suppose at least one dragon exists. Prove that the average of all terms which are the heads of some dragons is greater than 1988.

Olympiad Paper II.

1. (a) Suppose a_1, a_2 and a_3 are positive real numbers satisfying

$$(a_1^2 + a_2^2 + a_3^2)^2 > 2(a_1^4 + a_2^4 + a_3^4).$$

Prove that a_1, a_2 and a_3 are the three sides of some triangle.

- (b) Let n be an integer greater than 3. Suppose the inequality

$$(a_1^2 + a_2^2 + \cdots + a_n^2) > (n-1)(a_1^4 + a_2^4 + \cdots + a_n^4)$$

holds for some positive real numbers $a_1, a_2, \dots, a_n$. Prove that a_i, a_j and a_k are the three sides of some triangle for all i, j and k .

2. Given are three tetrahedra $A_i B_i C_i D_i$, $1 \leq i \leq 3$. Through the points B_i, C_i and D_i are drawn three planes β_i, γ_i , and δ_i perpendicular to $A_i B_i, A_i C_i$ and $A_i D_i$ respectively. If these nine planes meet at a point and A_1, A_2 and A_3 lie on a line, determine the intersection of the circumspheres of the three tetrahedra.
3. For any integer $n \geq 3$, let $f(n)$ denote the least positive integer which is not a divisor of n . Let $f^{(1)}(n) = f(n)$. If $f^{(k)}(n) \geq 3$, $f(f^{(k)}(n))$ is meaningful and is defined as $f^{(k+1)}(n)$. For each integer $n \geq 3$, determine k where $f^{(k)}(n) = 2$.

Olympiad Paper I.

1. Each of A and B is a union of pairwise disjoint arcs on the unit circle. Moreover, each arc in B is of length $\frac{\pi}{m}$, where m is a fixed positive integer. Denote by A^j the set obtained from A by rotating it about the origin through $\frac{j\pi}{m}$ radians in the counter-clockwise direction. Prove that there exists a positive integer k such that $\ell(A^k \cap B) \geq \frac{1}{2\pi} \ell(A) \ell(B)$, where $\ell(X)$ denotes the total length of a set X of disjoint arcs.

2. Suppose $x_1 + x_2 + \cdots + x_n = 1$ where $x_1, x_2, \dots, x_n$ are positive real numbers. Prove that

$$\frac{x_1}{\sqrt{1-x_1}} + \frac{x_2}{\sqrt{1-x_2}} + \cdots + \frac{x_n}{\sqrt{1-x_n}} \geq \frac{\sqrt{x_1} + \sqrt{x_2} + \cdots + \sqrt{x_n}}{\sqrt{n-1}}.$$

3. Let S be the unit circle in the complex plane. Let $f : S \rightarrow S$ be defined by $f(z) = z^m$, where m is a positive integer. Define $f^{(0)}(z) = z$ and $f^{(k+1)}(z) = f(f^{(k)}(z))$ for $k \geq 0$. The smallest positive integer n such that $f^{(n)}(z) = z$ is called the period of z . Determine the total number of points in S of period 1989.

Olympiad Paper II.

1. Triangle ABC has inradius r . D, E and F are points on BC, CA and AB respectively. If the inradii of triangles AEF, BFD and CDE are all equal, prove that their common value is $r - r'$, where r' is the inradius of triangle DEF .
2. Given are 1989 points in the plane, no 3 on a line. How should they be partitioned into 30 groups of distinct sizes so as to maximize the total number of triangles with vertices in different groups?
3. Determine all functions $f : S \rightarrow S$, where S is the set of all real numbers greater than 1, such that $f(x^m y^n) \leq f(x)^{\frac{1}{4m}} f(y)^{\frac{1}{4n}}$ for all real numbers $x, y > 1$ and $m, n > 0$.

Olympiad Paper I.

1. $ABCD$ is a convex quadrilateral with AB not parallel to CD . A circle passing through A and B is tangent to CD at P , and a circle passing through C and D is tangent to AB at Q . Prove that the common chord of these two circles bisects PQ if and only if AD is parallel to BC .
2. For a given positive integer x , a D -chain of x of length d is defined to be a sequence

$$1 = x_0 < x_1 < x_2 < \cdots < x_d = x$$

such that x_i divides x_{i+1} for $1 \leq i \leq d-1$. For $x = 5^k 31^m 1990^n$ where k, m and n are positive integers, find

- (a) the maximum length of a D -chain;
- (b) the number of D -chains of this length.

3. A real-valued function f defined for all non-negative real numbers is such that

$$f(x)f(y) \leq y^2 f\left(\frac{x}{2}\right) + x^2 f\left(\frac{y}{2}\right)$$

for all $x \geq y \geq 0$ and $|f(x)| \leq M$ for $0 \leq x \leq 1$, where M is a fixed constant. Prove that $f(x) \leq x^2$.

Olympiad Paper II.

1. Let a be a positive integer and A and B be real numbers. Consider the system of equations $x^2(Ax^2 + By^2) + y^2(Ay^2 + Bz^2) + z^2(Az^2 + Bx^2) = \frac{1}{4}(2A + B)(13a)^4$ and $x^2 + y^2 + z^2 = (13a)^2$. Find a necessary and sufficient condition on A and B for this system to have positive integer solutions in x, y and z .
2. Let X be a finite set and $E(X)$ be the collection of subsets of X with an even number of elements. A real-valued function f is defined on $E(X)$ such that $f(D) > 1990$ for at least one D in $E(X)$ and $f(A \cup B) = f(A) + f(B) - 1990$ for all disjoint A and B in $E(X)$. Prove that X may be partitioned into two disjoint subsets P and Q such that $f(S) > 1990$ for all non-empty S in $E(P)$ and $f(T) \leq 1990$ for all T in $E(Q)$.
3. A convex n -gon may be partitioned into $n-2$ triangles by $n-3$ of its diagonals with no two intersecting except at the vertices. Prove that all sides and diagonals of at least one such partitions can be drawn without repetition as a continuous closed polygonal path if and only if n is a multiple of 3.

Olympiad Paper I. 1991/1/12 8:00-12:30

1. (a) What conditions must a convex quadrilateral $ABCD$ satisfy if there exists a point P on the plane of $ABCD$ such that the areas of triangles ABP , BCP , CDP and DAP are all equal?
 (b) At most how many such points P can there be for a given $ABCD$?

2. Find all functions $f : [0, 1] \times [0, 1] \rightarrow [0, 1]$ such that for all x, y and z in $[0, 1]$,

$$\begin{aligned} f(x, 1) &= x, \\ f(1, y) &= y, \\ f(f(x, y), z) &= f(x, f(y, z)) \\ \text{and } f(zx, zy) &= z^k f(x, y) \end{aligned}$$

for some positive constant k independent of x, y and z .

3. Ten little birds are pecking grains on a level surface. For any five birds, at least four of them are on the circumference of some circle. What is the minimum value of the maximum number of these ten birds on the circumference of a circle?

Olympiad Paper II. 1991/1/13 8:00-12:30

1. Find all quadruples (n, x, y, z) of positive integers which satisfy $n \geq 2$, $z \leq 5 \cdot 2^{2n}$ and

$$x^{2n+1} - y^{2n+1} = xyz + 2^{2n+1}.$$

2. Find all positive integers n such that 1991 is the minimum value of

$$k^2 + \left\lfloor \frac{n}{k^2} \right\rfloor,$$

where k ranges over all positive integers.

3. Every vertex of a certain polyhedron lies on exactly three edges, and it is possible to colour the edges so that each vertex lies on one edge of each colour. Prove that it is possible to assign to each vertex a complex number not equal to 1, such that the product of the numbers on the vertices of each face is equal to 1.

Olympiad Paper I.

1. Let $0 < a_0 \leq a_1 \leq \cdots \leq a_{n-1} \leq 1$ be real numbers. Let λ be a complex root of the equation $x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$ such that $|\lambda| \geq 1$. Prove that $\lambda^{n+1} = 1$.

2. Let $x_1, x_2, \dots, x_n$ be non-negative real numbers and let a be the minimum of these numbers.

(a) Prove that

$$\begin{aligned} & \frac{1+x_1}{1+x_2} + \frac{1+x_2}{1+x_3} + \dots + \frac{1+x_{n-1}}{1+x_n} + \frac{1+x_n}{1+x_1} \\ & \leq n + \frac{(x_1-a)^2 + (x_2-a)^2 + \dots + (x_n-a)^2}{(1+a)^2}. \end{aligned}$$

(b) Prove that equality holds if and only if $x_1 = x_2 = \dots = x_n$.

3. Initially, the number 1 or -1 is arbitrarily assigned to each cell of a 9×9 board. For each cell C , compute the product of all numbers in cells sharing exactly one edge with C . Simultaneously, all numbers are replaced by the corresponding products. Is it always possible, by performing the above operation a finite number of times, to change all 81 numbers to 1's?

Olympiad Paper II.

1. A convex quadrilateral $ABCD$ is inscribed in a circle with centre O . The diagonals AC and BD intersect at P . The circumcircles of triangles ABP and CDP intersect again at Q . If O, P and Q are three distinct points, prove that OQ is perpendicular to PQ .
2. A graph with 8 vertices has no loops, multiple edges or 4-cycles. At most how many edges can it have?
3. Let a_0 and a_1 be integers. The sequence a_n is defined by $a_2 = 2a_1 - a_0 + 2$ and

$$a_{n+1} = 3a_n - 3a_{n-1} + a_{n-2}$$

for $n \geq 2$. If for any positive integer m , the sequence contains m consecutive terms all of which are perfect squares, prove that every term of the sequence is a perfect square.

Olympiad Paper I.

1. Let n be an odd positive integer. Prove that there exist $2n$ integers $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$, such that for any integer $k, 0 < k < n$, the following $3n$ integers are pairwise non-congruent modulo $3n$: $a_i + a_{i+1}, a_i + b_i$ and $a_i + b_{i+k}, 1 \leq i \leq n$, where the subscripts are reduced modulo n .

2. Let n be a positive integer and a a positive real number. Determine the maximum value of $a^{k(1)} + a^{k(2)} + \dots + a^{k(s)}$, where s is an integer such that $1 \leq s \leq n$ and $k(1), k(2), \dots, k(s)$ are positive integers with sum n .

3. Two concentric circles have radii R and R_1 respectively, where $R_1 > R$. $ABCD$ is inscribed in the smaller circle and $A_1B_1C_1D_1$ in the larger one, with A_1 on the extension of CD , B_1 on that of DA , C_1 on that of AB and D_1 on that of BC . Prove that

$$\frac{[A_1B_1C_1D_1]}{[ABCD]} \geq \frac{R_1^2}{R^2},$$

where $[P]$ denotes the area of the polygon P .

Olympiad Paper II.

1. Let S be a set of 1993 non-zero vectors in the plane. Prove that S has a collection of non-empty subsets with the following properties:
 - Every vector in S belongs to exactly one of the subsets.
 - The angle between any vector in a subset and the resultant vector of that subset is at most 90° .
 - The angle between the resultant vectors of any two subsets is greater than 90° .
2. Ten people went to buy books. Each bought three different books. Every two bought at least one book in common. Consider the book bought by the highest number of these ten people. What is the minimum value of this number?
3. Let f be a function from the positive real numbers to the positive real numbers such that $f(xy) \leq f(x)f(y)$ for any positive real numbers x and y . Prove that for any positive real number x and all positive integers n ,

$$f(x^n) \leq f(x)f(x^2)^{\frac{1}{2}} \cdots f(x^n)^{\frac{1}{n}}.$$